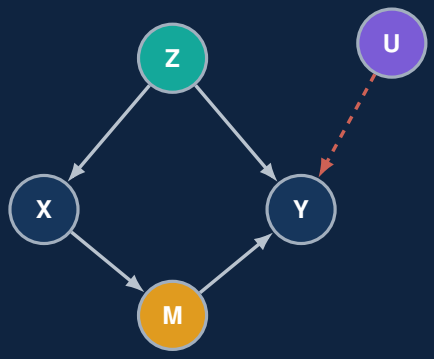



ADVANCED ARTIFICIAL INTELLIGENCE

# Probabilistic Graphical Models and Causal Inference

## Session 14

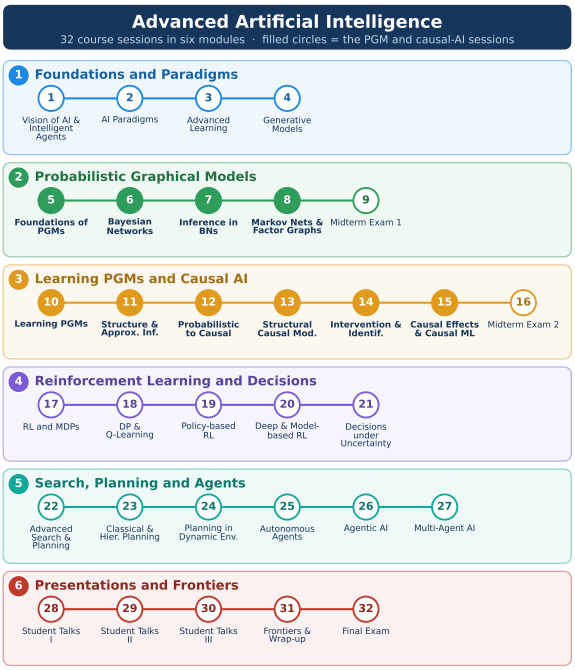
## Intervention and Identification: Do-Calculus

Lecture Notes

Instructor: Prof. M. Arashi

# About these notes

These notes cover Session 14 of the course *Advanced Artificial Intelligence*, part of its probabilistic-graphical-model and causal-AI block (course sessions 5–15). The session follows a fixed structure: a roadmap and learning objectives, the core theory organized in parts, worked numerical examples, and a part connecting the topic to intelligent agents, agentic AI, reinforcement learning, multi-agent systems and the other sessions of the course. Every number is reproduced by the companion Python scripts (`python/s14_examples.py` for the numbers, `python/s14_figures.py` for the figures), and `questions/s14_questions.tex` contains ten questions with complete answers.




The Advanced Artificial Intelligence course. Filled circles mark the PGM and causal-AI sessions; this is session 14.

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SESSION 14 ADVANCED ARTIFICIAL INTELLIGENCE

# Intervention and Identification: Do-Calculus

The do-operator, do-calculus, back-door and front-door criteria, and identification

Advanced Artificial Intelligence · Probabilistic Graphical Models and Causal Inference

Instructor: Prof. M. Arashi

**PART I**
**Interventions**
The do-operator, the identification problem, policies and soft interventions.

**PART II**
**Back-door & Front-door**
Adjustment, IPW, the kidney data, smoking–tar–cancer.

**PART III**
**Do-Calculus & ID**
Three rules, instruments, c-components, non-identifiability.

**PART IV**
**AI Connections**
RL actions, off-policy logs, tool-using agents, interacting agents.

## LEARNING OBJECTIVES

- Define $P(y \mid \mathrm{do}(x))$, policy interventions $\mathrm{do}(\pi)$ and identifiability, and explain why identification depends on the graph, not on the sample size.
- State and prove the back-door adjustment formula and its IPW form; derive the front-door formula with the three rules of do-calculus.
- Go beyond both criteria with instruments, c-components and the ID algorithm; prove non-identifiability of the bow arc by constructing two SCMs.
- Turn a causal question into an estimand, with bounds as a fall-back.
- Recognise identification problems in RL, in tool-using LLM agents and in multi-agent systems.

**Where we are.** Session 12 showed that $P(y \mid x)$ can have the wrong sign as an answer to "what if we act?"; Session 13 built structural causal models (SCMs), in which $\mathrm{do}(X = x)$ is graph surgery computed *from the mechanisms*. This session (course session 14) drops that luxury: we know only a graph and observational data, and ask when $P(y \mid \mathrm{do}(x))$ is still computable. Session 15 estimates the identified quantities with machine learning; reinforcement learning (RL, course sessions 17–20), agentic AI (course session 26) and multi-agent AI (course session 27) need today's results whenever an agent learns about actions from data it did not generate.

## OPENING MOTIVATION

Session 13's truncated factorization formula computes $P(y \mid \mathrm{do}(x))$ — *but only if we already have the full set of structural equations*, confounders and all. In practice we almost never

observe every confounder: no dataset records "ability" directly, and no observational cancer study directly measures every genetic predisposition. The practitioner's real question is sharper: given only an observational dataset and a *graph* (which may include boxes for variables we know exist causally but cannot measure), can we still write $P(y \mid \mathrm{do}(x))$ as a formula using only observable quantities? This is the *identification* problem, and today's answer — the back-door criterion, the front-door criterion, and the complete rules of do-calculus — turns Session 12's confounding heuristics into checkable theorems.

PART I

# Interventions and the Identification Problem

What $\mathrm{do}(\cdot)$ means, what we may assume, and what we want to compute

PART I · INTERVENTIONS

## 14.1 The do-Operator Revisited

Let $M$ be a Markovian SCM with causal DAG $G$ over observed variables $\mathbf{V}$ (Session 13). The intervention $\mathrm{do}(X = x)$ replaces the mechanism of $X$ by the constant $x$, producing the submodel $M_x$ whose graph $G_{\overline{X}}$ has all arrows *into* $X$ deleted. Its distribution is given by the truncated factorization

$$P(\mathbf{v} \mid \mathrm{do}(x)) = \prod_{i:\, V_i \notin X} P(v_i \mid \mathrm{pa}_i)\Big|_{X=x} \quad \text{for } \mathbf{v} \text{ consistent with } x, \qquad 0 \text{ otherwise.} \tag{14.1}$$

Throughout, we use the graph-surgery notation of Table 14.1.

**Table 14.1.** Mutilated graphs used by the back-door criterion and do-calculus.

| Graph | Edges removed | Used to check |
|---|---|---|
| $G_{\overline{X}}$ | all arrows *into* $X$ | the world after $\mathrm{do}(x)$; Rule 1 |
| $G_{\underline{X}}$ | all arrows *out of* $X$ | only back-door paths of $X$ remain; Rule 2, back-door criterion |
| $G_{\overline{X}\underline{Z}}$ | into $X$ and out of $Z$ | exchanging $\mathrm{do}(z)$ and $z$ (Rule 2) |
| $G_{\overline{X}\,\overline{Z(W)}}$ | into $X$, and into the $Z$-nodes that are not ancestors of $W$ | deleting $\mathrm{do}(z)$ (Rule 3) |

For the confounded triangle $Z \to X$, $Z \to Y$, $X \to Y$, Eq. (14.1) with $\mathbf{V} = \{Z, X, Y\}$ gives $P(z, y \mid \mathrm{do}(x)) = P(z)P(y \mid x, z)$, hence $P(y \mid \mathrm{do}(x)) = \sum_z P(y \mid x, z)P(z)$, while the law of total probability gives $P(y \mid x) = \sum_z P(y \mid x, z)P(z \mid x)$. Seeing reweights the strata by $P(z \mid x)$; doing keeps $P(z)$. Both are computable — *because* $Z$ was observed. This session is about what happens when it is not.

PART I · INTERVENTIONS

## 14.2 The Identification Problem

**Definition 14.1** (Identifiability)**.** A causal effect $P(y \mid \mathrm{do}(x))$ is *identifiable* from a causal graph $G$ if it can be expressed as a function of the observational distribution $P(\mathbf{v})$ over the observed variables alone — computable without knowledge of the specific structural equations or unobserved-variable distributions, using only $G$'s structure and observational data.

Formally: for every pair of SCMs $M_1, M_2$ that induce the graph $G$ and satisfy $P_{M_1}(\mathbf{v}) = P_{M_2}(\mathbf{v}) > 0$, we require $P_{M_1}(y \mid \mathrm{do}(x)) = P_{M_2}(y \mid \mathrm{do}(x))$. Three consequences follow. (i) Identification concerns the *population* ("could infinite data reveal it?"); more data never rescues a non-identified effect, which is not estimated with large variance but undefined by the data. (ii) To *disprove* it, exhibit two models that agree on $P(\mathbf{v})$ and disagree on $P(y \mid \mathrm{do}(x))$ (Section 14.9). (iii) Positivity matters: a formula dividing by $P(x \mid z)$ is meaningless where $X = x$ never occurs.

PART I · INTERVENTIONS

## 14.3 Interventions Beyond do(x): Policies and Soft Interventions

Real decisions are rarely atomic: a doctor treats by severity, a recommender randomizes, an RL agent follows a stochastic policy. Pearl (2009, Ch. 4) treats such *policy* interventions by the same surgery.

**Definition 14.2** (Conditional, stochastic and soft interventions)**.** Let $Z$ be a set of non-descendants of $X$. A *conditional intervention* $\mathrm{do}(X = g(Z))$ replaces the mechanism of $X$ by the deterministic rule $x = g(z)$. A *stochastic policy* $\mathrm{do}(\pi)$ replaces it by a fresh random draw $X \sim \pi(\cdot \mid Z)$, independent of all other exogenous noise given $Z$. More generally, a *soft* (mechanism-changing) intervention replaces $P(x \mid \mathrm{pa}_X)$ by any new kernel, possibly with new parents. Atomic interventions are the special case $\pi(x' \mid z) = \mathbb{1}(x' = x)$.

**Proposition 14.1** (Value of a policy)**.** *If $Z$ consists of non-descendants of $X$ and satisfies the back-door criterion (Definition 14.4) relative to $(X, Y)$, then for every policy $\pi(x \mid z)$*

$$P\big(y \mid \mathrm{do}(\pi)\big) = \sum_{z} \sum_{x} P(y \mid x, z)\, \pi(x \mid z)\, P(z). \tag{14.2}$$

*Proof.* Let $M_\pi$ be the SCM in which $X := h(Z, \varepsilon)$ with $P(h(z, \varepsilon) = x) = \pi(x \mid z)$ and $\varepsilon$ independent of all original noises. Then $P_\pi(y) = \sum_{z,x} P_\pi(z) P_\pi(x \mid z) P_\pi(y \mid x, z)$. (a) The mechanisms of the non-descendants of $X$ and their noises are untouched, so $P_\pi(z) = P(z)$. (b) $P_\pi(x \mid z) = \pi(x \mid z)$ by construction. (c) Given $Z = z$, the event $\{X = x\}$ in $M_\pi$ is determined by $\varepsilon$, which is independent of everything else; conditioning on it is therefore the same as setting $X = x$, and $P_\pi(y \mid x, z) = P(y \mid \mathrm{do}(x), z)$. Finally $P(y \mid \mathrm{do}(x), z) = P(y \mid x, z)$ by Rule 2 of do-calculus, because $Z$ blocks every back-door path (Section 14.6). □

**Table 14.2.** The real kidney-stone data (Charig et al. 1986; Session 12): recovered/treated patients. Treatment A = open surgery, B = percutaneous nephrolithotomy (PCNL); $Z$ = stone size.

| Stone size $Z$ | Treatment A | Treatment B | All patients |
|---|---|---|---|
| **small** | $81/87 = 0.9310$ | $234/270 = 0.8667$ | $357$ $(P(\text{small}) = 357/700 = 0.51)$ |
| **large** | $192/263 = 0.7300$ | $55/80 = 0.6875$ | $343$ $(P(\text{large}) = 343/700 = 0.49)$ |
| **pooled** | $273/350 = 0.7800$ | $289/350 = 0.8257$ | $562/700 = 0.8029$ recovered |

PART II

# The Back-door and Front-door Criteria

Two graphical certificates that turn $P(y \mid \text{do}(x))$ into observational quantities

PART II · ADJUSTMENT

## 14.4 The Back-door Criterion

**Definition 14.3** (Back-door path)**.** A path from $X$ to $Y$ is a *back-door path* if its first edge points *into* $X$, i.e. it has the form $X \leftarrow \cdots Y$. (Paths whose first edge is $X \rightarrow \cdots$ are not back-door paths, whether or not they are directed.)

**Definition 14.4** (Back-door criterion)**.** A set of variables $Z$ satisfies the back-door criterion relative to $(X, Y)$ if: (i) no node in $Z$ is a descendant of $X$, and (ii) $Z$ blocks every back-door path between $X$ and $Y$ (in the sense of d-separation, Session 6).

Condition (ii) is a single d-separation statement: $Z$ blocks all back-door paths iff $X \perp\!\!\!\perp_d Y \mid Z$ in $G_{\underline{X}}$, since removing the arrows out of $X$ leaves exactly the back-door paths. Checking the criterion is therefore as cheap as one Bayes-ball run (linear in the size of the graph, Session 6).

**Theorem 14.1** (Back-door adjustment)**.** *If $Z$ satisfies the back-door criterion relative to $(X, Y)$, then $P(y \mid \text{do}(x))$ is identifiable and equals*

$$P\big(y \mid \text{do}(x)\big) = \sum_{z} P(y \mid x, z)\, P(z). \tag{14.3}$$

*Proof for $Z = \text{pa}(X)$, from the truncated factorization.* Write $T = \text{pa}(X)$ and let $R = \mathbf{V} \backslash (\{X, Y\} \cup T)$. On configurations with $X = x$, Eq. (14.1) and the Markov factorization $P(\mathbf{v}) = P(x \mid t) \prod_{i \neq X} P(v_i \mid \text{pa}_i)$ give, wherever $P(x \mid t) > 0$,

$$P(\mathbf{v} \mid \text{do}(x)) = \frac{P(\mathbf{v})}{P(x \mid t)}.$$

Summing over $r$ and $t$: $P(y \mid \text{do}(x)) = \sum_t \sum_r P(x, y, t, r)/P(x \mid t) = \sum_t P(x, y, t)/P(x \mid t) = \sum_t P(y \mid x, t)P(t)$, which is Eq. (14.3) with $Z = T$. The parents of $X$ always satisfy the back-door

criterion, so this is the "adjustment for direct causes" special case. For a general back-door set $Z$ (which may exclude unobserved parents of $X$) the result follows from do-calculus; the derivation is given after the do-calculus rules in Section 14.6. □

This is exactly the stratify-and-reweight calculation Session 12's real-world investigators performed by hand on the Berkeley and kidney-stone data — now justified as the unique correct formula whenever $Z$ satisfies a checkable graphical condition, rather than a statistical instinct to "control for confounders."

**Worked Example 14.1** (Formally resolving the kidney-stone paradox)**.** In the SCM $X := g(Z, U_X)$, $Y := h(X, Z, U_Y)$ from Session 13, stone severity $Z$ satisfies the back-door criterion for $(X, Y)$: it is not a descendant of treatment $X$ (severity is determined before treatment is chosen), and it blocks the only back-door path $X \leftarrow Z \rightarrow Y$. Applying Theorem 14.1 to the *real* data from Session 12 (Table 14.2): $P(Z{=}\text{small}) = (87 + 270)/700 = 357/700 = 0.510$, $P(Z{=}\text{large}) = (263 + 80)/700 = 343/700 = 0.490$, and the stratum recovery rates are $81/87 = 0.9310$, $192/263 = 0.7300$ for A and $234/270 = 0.8667$, $55/80 = 0.6875$ for B. So

$$P(\text{recover} \mid \text{do}(A)) = 0.9310(0.510) + 0.7300(0.490) = 0.4748 + 0.3577 = 0.8325,$$
$$P(\text{recover} \mid \text{do}(B)) = 0.8667(0.510) + 0.6875(0.490) = 0.4420 + 0.3369 = 0.7789.$$

Treatment A's interventional recovery probability (0.8325) now correctly exceeds Treatment B's (0.7789), matching what both size-stratified subgroups already showed individually in Session 12. The naive pooled rates $P(\text{recover} \mid A) = 273/350 = 0.7800$ and $P(\text{recover} \mid B) = 289/350 = 0.8257$ weight the strata by $P(z \mid t)$ instead of $P(z)$: $P(\text{large} \mid A) = 263/350 = 0.7514$ but $P(\text{large} \mid B) = 80/350 = 0.2286$, so A is charged with the hard cases. Theorem 14.1 is precisely the formula that makes re-weighting by $P(z)$ the mathematically *correct* way to combine the two strata. The *average treatment effect* (ATE) of A versus B is $P(\text{recover} \mid \text{do}(A)) - P(\text{recover} \mid \text{do}(B)) = 0.83255 - 0.77887 = +0.0537$ (difference of the unrounded values), against the naive $0.7800 - 0.8257 = -0.0457$.

**PYTHON LAB** · s14_examples.py · ex_kidney_backdoor()

```
P(Z=small) = 357/700 = 0.510; naive P(rec|A) = 273/350 = 0.7800, P(rec|B) =
289/350 = 0.8257
P(rec|do(A)) = 0.4748 + 0.3577 = 0.8325, P(rec|do(B)) = 0.4420 + 0.3369 =
0.7789
P(large|A) = 0.7514, P(large|B) = 0.2286; naive A-B = -0.0457, adjusted ATE
= 0.83255 - 0.77887 = +0.0537
```

### 14.4.1 Several back-door sets: which to choose?

All valid sets give the *same* population value $P(y \mid \text{do}(x))$; they differ in finite-sample variance (Session 15).

### 14.4.2 Adjustment as weighting: the IPW representation

*Inverse-probability weighting* (IPW) rewrites the adjustment formula as a weighted average over units, with the *propensity* $P(x \mid z)$ in the denominator.

**Proposition 14.2** (IPW form of the adjustment formula)**.** *Let $Z$ satisfy the back-door criterion and suppose positivity, $P(x \mid z) > 0$ whenever $P(z) > 0$. Then*

$$P(y \mid \mathrm{do}(x)) = \mathbb{E}\left[\frac{\mathbb{1}(X = x)\mathbb{1}(Y = y)}{P(x \mid Z)}\right], \qquad \mathbb{E}[Y \mid \mathrm{do}(x)] = \mathbb{E}\left[\frac{\mathbb{1}(X = x)\, Y}{P(x \mid Z)}\right]. \tag{14.4}$$

*Proof.* Condition on $Z$: $\mathbb{E}\big[\mathbb{1}(X{=}x)\mathbb{1}(Y{=}y)/P(x \mid Z)\,\big|\,Z = z\big] = P(X{=}x, Y{=}y \mid z)/P(x \mid z) = P(y \mid x, z)$. Taking the expectation over $Z$ gives $\sum_z P(y \mid x, z)P(z)$, which is Eq. (14.3). The second identity follows by summing $y\,P(y \mid \mathrm{do}(x))$ over $y$. □

Adjustment *models the outcome*; IPW *models the treatment*, reweighting units by $1/P(x \mid z)$ into a pseudo-population in which $X \perp\!\!\!\perp Z$ — a simulated randomized trial. Session 15 combines both (doubly robust estimation).

### 14.4.3 Linear models: adjustment is multiple regression

In a linear SEM, $\mathbb{E}[Y \mid \mathrm{do}(x), z]$ is linear in $(x, z)$, so Eq. (14.3) says: regress $Y$ on $X$ *and* a back-door set $Z$ and read off the coefficient of $X$. Adding the wrong variable is dangerous:

**Worked Example 14.2** (Over-control: adjusting for a mediator)**.** *(Analytical, with a simulation check.)* Consider the linear SEM with a mediator $W$: $Z = U_Z$, $X = 0.8Z + U_X$, $W = X + U_W$, $Y = 1.5W + 2Z + U_Y$, where $U_Z, U_X, U_W, U_Y$ are independent $N(0, 1)$ and there is no direct $X \to Y$ arrow. The total effect is $\partial\mathbb{E}[Y \mid \mathrm{do}(x)]/\partial x = 1 \times 1.5 = 1.5$ (the product of the coefficients along $X \to W \to Y$), and $\{Z\}$ is a back-door set. *Regression on $(X, Z)$:* substituting $W$, $Y = 1.5X + 2Z + (1.5U_W + U_Y)$ with an error independent of $(X, Z)$, so the population coefficients are (1.5, 2) — correct. *Regression on $(X, Z, W)$:* $Y = 0 \cdot X + 2Z + 1.5W + U_Y$ with $U_Y$ independent of $(X, Z, W)$, so these are already the population normal equations and the coefficients are (0, 2, 1.5). Adjusting for the descendant $W$ removes the *entire* effect — the coefficient of $X$ is 0, not 1.5. This is condition (i) of the back-door criterion at work. Simulation (`ex_overcontrol()`, $n = 2 \times 10^5$, seed 15): 1.500 with $\{Z\}$; −0.002 with $\{Z, W\}$.

**PYTHON LAB** · `s14_examples.py` · `ex_overcontrol()`

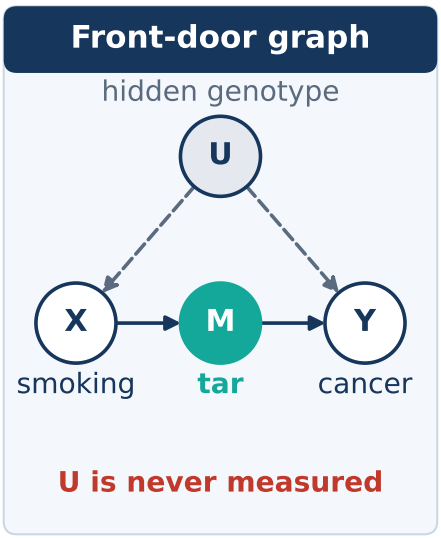
```
simulation n=200000, seed=15:  Y   X + Z : coef on X = 1.500
Y   X + Z + W : coef on X = -0.002, on Z = 2.000, on W = 1.502
```

PART II · ADJUSTMENT

## 14.5 The Front-Door Criterion: When Adjustment Is Impossible but Identification Still Succeeds

The back-door criterion requires *measuring* enough of the confounding structure to block every back-door path. What if the confounder itself is unmeasurable?

**Worked Example 14.3** (Smoking, tar, and cancer)**.** Consider $X =$ smoking, $Y =$ lung cancer, with a hypothesized unmeasured genetic factor $U$ that predisposes people both to smoke and (independently) to develop cancer — a classic historical objection to the smoking-cancer causal claim. Suppose additionally that smoking's effect on cancer is entirely mediated by tar deposits in the lungs, $M$: $X \to M \to Y$, with $U$ affecting $X$ and $Y$ directly but *not* $M$ (Fig. 14.1).




**1** **X → M** **no open back-door X–M**

$$P(m \mid do(x)) = P(m \mid x)$$

**2** **M → Y** **X blocks M ← X ← U → Y**

$$P(y \mid do(m)) = \sum_{x'} P(y \mid m, x')\, P(x')$$

**3** **chain** **M carries all of the effect**

$$P(y \mid do(x)) = \sum_{m} P(m \mid do(x))\, P(y \mid do(m))$$

**Figure 14.1.** The front-door configuration: $X \to Y$ is confounded by unobserved $U$, but the mediating path through $M$ is unconfounded in both legs. Right: the three do-calculus steps of Theorem 14.2's derivation.

**Definition 14.5** (Front-door criterion)**.** A set of variables $M$ satisfies the front-door criterion relative to $(X, Y)$ if: (i) $M$ intercepts all directed paths from $X$ to $Y$; (ii) there is no unblocked back-door path from $X$ to $M$; (iii) every back-door path from $M$ to $Y$ is blocked by $X$.

**Theorem 14.2** (Front-door adjustment)**.** *If $M$ satisfies the front-door criterion relative to $(X, Y)$ and $P(x, m) > 0$, then $P(y \mid \mathrm{do}(x))$ is identifiable and equals*

$$P(y \mid \mathrm{do}(x)) = \sum_{m} P(m \mid x) \sum_{x'} P(y \mid x', m)\, P(x'). \tag{14.5}$$

*Derivation by do-calculus (graph of Fig. 14.1).* **1** *Effect of $X$ on $M$.* In $G_{\underline{X}}$ the only path between $X$ and $M$ is $X \leftarrow U \to Y \leftarrow M$, blocked at the collider $Y$; so $M \perp\!\!\!\perp_d X$ in $G_{\underline{X}}$ and Rule 2 gives $P(m \mid \mathrm{do}(x)) = P(m \mid x)$.

**2** *Effect of $M$ on $Y$.* Condition on $X$: $P(y \mid \mathrm{do}(m)) = \sum_{x'} P(y \mid \mathrm{do}(m), x') P(x' \mid \mathrm{do}(m))$. In $G_{\overline{M}}$ the arrow $X \to M$ is cut and the only $X$–$M$ path $X \leftarrow U \to Y \leftarrow M$ is blocked at $Y$, so Rule 3 gives $P(x' \mid \mathrm{do}(m)) = P(x')$. In $G_{\underline{M}}$ the remaining $M$–$Y$ path $M \leftarrow X \leftarrow U \to Y$ is blocked by conditioning on $X$, so Rule 2 gives $P(y \mid \mathrm{do}(m), x') = P(y \mid m, x')$. Hence $P(y \mid \mathrm{do}(m)) = \sum_{x'} P(y \mid m, x') P(x')$.

**3** *Chain the legs.* $P(y \mid \mathrm{do}(x)) = \sum_{m} P(y \mid \mathrm{do}(x), m) P(m \mid \mathrm{do}(x))$. In $G_{\overline{X}\underline{M}}$ ($U \to X$ and $M \to Y$ cut) $X$ is attached only to $M$, so $Y \perp\!\!\!\perp_d M \mid X$ and Rule 2 gives $P(y \mid \mathrm{do}(x), m) = P(y \mid \mathrm{do}(x), \mathrm{do}(m))$; in $G_{\overline{X}\,\overline{M}}$ the node $X$ is isolated, so Rule 3 gives $P(y \mid \mathrm{do}(x), \mathrm{do}(m)) = P(y \mid \mathrm{do}(m))$. Substituting steps 1 and 2 yields Eq. (14.5). □

**INTUITION**

The formula estimates the effect in two unconfounded legs and chains them: $P(m \mid x)$ is the (unconfounded, by condition ii) effect of smoking on tar; the inner sum $\sum_{x'} P(y \mid x', m)P(x')$ estimates the (unconfounded once we further adjust for $X$, by condition iii) effect of tar on cancer, holding the confounding influence of $U$ on $Y$ fixed by averaging over $X$'s marginal distribution rather than conditioning on the specific $x$ we started with. The genuinely surprising part is that the unmeasured, un-adjustable confounder $U$ between $X$ and $Y$ never needs to be measured at all — it is defeated by routing around it through a mediator both of whose legs happen to be individually unconfounded.

**Worked Example 14.4** (Front-door numerically)**.** Suppose (illustrative numbers) $P(M{=}1 \mid X{=}1) = 0.8$, $P(M{=}1 \mid X{=}0) = 0.2$ (smoking strongly increases tar), $P(X{=}1) = 0.4$ in the population, and $P(Y{=}1 \mid X{=}1, M{=}1) = 0.5$, $P(Y{=}1 \mid X{=}0, M{=}1) = 0.5$, $P(Y{=}1 \mid X{=}1, M{=}0) = 0.1$, $P(Y{=}1 \mid X{=}0, M{=}0) = 0.1$ (in these illustrative numbers cancer risk depends only on tar: $P(y \mid x, m)$ does not vary with $x$). Then $\sum_{x'} P(Y{=}1 \mid x', M{=}1)P(x') = 0.5(0.4) + 0.5(0.6) = 0.5$ and similarly $\sum_{x'} P(Y{=}1 \mid x', M{=}0)P(x') = 0.1$. So $P(Y{=}1 \mid \mathrm{do}(X{=}1)) = P(M{=}1 \mid X{=}1)(0.5) + P(M{=}0 \mid X{=}1)(0.1) = 0.8(0.5) + 0.2(0.1) = 0.42$, versus $P(Y{=}1 \mid \mathrm{do}(X{=}0)) = 0.2(0.5) + 0.8(0.1) = 0.18$ — a causal effect of smoking on cancer of $0.42 - 0.18 = 0.24$, computed without measuring $U$. Note, however, that here the naive contrast is the same: $P(Y{=}1 \mid X{=}1) = 0.8(0.5) + 0.2(0.1) = 0.42$ and $P(Y{=}1 \mid X{=}0) = 0.2(0.5) + 0.8(0.1) = 0.18$. Because $P(y \mid x, m)$ does not depend on $x$, these particular numbers carry no trace of confounding by $U$ (given $M$, $X$ tells us nothing about $Y$, hence nothing about $U$'s effect on $Y$). The next example has genuine confounding.

**PYTHON LAB** · `s14_examples.py` · `ex_frontdoor_numeric()`

```
inner sums:  m=1 -> 0.50, m=0 -> 0.10; P(Y=1|do(X=1)) = 0.42, P(Y=1|do(X=0))
= 0.18, effect = 0.24
```

PART III

# Do-Calculus and Identification

Three rules, instruments, c-components, and the limits of identification

PART III · DO-CALCULUS

## 14.6 Do-Calculus: The Complete Rules

Back-door and front-door are the two most commonly invoked special cases of a fully general calculus for manipulating do(·) expressions, built entirely from d-separation applied to mutilated graphs. Write $G_{\overline{X}}$ for $G$ with $X$'s incoming edges removed (Session 13's mutilation) and $G_{\underline{X}}$ for $G$ with $X$'s outgoing edges removed.

**Theorem 14.3** (Rules of do-calculus, Pearl 1995)**.** *For disjoint variable sets $X, Y, Z, W$:*

**Rule 1 (insertion/deletion of observations):** $P(y \mid \mathrm{do}(x), z, w) = P(y \mid \mathrm{do}(x), w)$ *if* $(Y \perp\!\!\!\perp_d Z \mid X, W)$ *in* $G_{\overline{X}}$.

**Rule 2 (action/observation exchange):** $P(y \mid \mathrm{do}(x), \mathrm{do}(z), w) = P(y \mid \mathrm{do}(x), z, w)$ *if* $(Y \perp\!\!\!\perp_d Z \mid X, W)$ *in* $G_{\overline{X}\underline{Z}}$.

**Rule 3 (insertion/deletion of actions):** $P(y \mid \mathrm{do}(x), \mathrm{do}(z), w) = P(y \mid \mathrm{do}(x), w)$ *if* $(Y \perp\!\!\!\perp_d Z \mid X, W)$ *in* $G_{\overline{X}\,\overline{Z(W)}}$, *where* $Z(W)$ *excludes any* $Z$*-nodes that are ancestors of* $W$ *in* $G_{\overline{X}}$.

*These three rules are* sound and complete*: a causal effect is identifiable if and only if it can be reduced to a purely observational expression by some finite sequence of applications of Rules 1–3 (together with ordinary probability axioms).*

We state completeness without proof (it is a substantial graph-theoretic result; Shpitser & Pearl, 2006, give an algorithm that always finds the reduction when one exists). What matters for practice is the intuition behind each rule: every one of them is exactly Session 6's d-separation criterion, applied not to the original graph but to a graph mutilated by the actions already performed — do-calculus adds nothing conceptually new to d-separation, it only asks us to check separation in the right (possibly surgically modified) graph for each manipulation we want to justify. Both Theorem 14.1 and Theorem 14.2 can be derived as short sequences of these three rules; we did this for the front-door case above and do it for the back-door case in the appendix.

**Why the rules are sound (sketch).** Give each intervened variable $Z$ a "switch" parent $F_Z$ (idle or a forced value), so that $P(y \mid \mathrm{do}(z), \cdot) = P(y \mid F_Z{=}z, \cdot)$; each rule is then an ordinary d-separation statement in the augmented DAG (Rule 2: $F_Z$ is irrelevant once $Z$ is observed; Rule 3: $F_Z$ is irrelevant altogether), and soundness follows from that of d-separation and the causal Markov condition.

**The back-door formula from the rules.** For $Z$ satisfying the back-door criterion, start from $P(y \mid \mathrm{do}(x)) = \sum_z P(y \mid \mathrm{do}(x), z) P(z \mid \mathrm{do}(x))$. *Rule 3* gives $P(z \mid \mathrm{do}(x)) = P(z)$, because no node of $Z$ is a descendant of $X$, so $Z \perp\!\!\!\perp_d X$ in $G_{\overline{X}}$ (every path from $X$ in $G_{\overline{X}}$ starts with an outgoing arrow and must meet a collider before reaching a non-descendant). *Rule 2* gives $P(y \mid \mathrm{do}(x), z) = P(y \mid x, z)$, because $Z$ blocks every back-door path, i.e. $Y \perp\!\!\!\perp_d X \mid Z$ in $G_{\underline{X}}$. Substituting yields Theorem 14.1.

**Worked Example 14.5** (An explicit application of Rule 1)**.** Let $X \to Y$ and also $X \to W$, with no edge between $W$ and $Y$ at all ($W$ is an unrelated side-effect of $X$). Show $P(y \mid \mathrm{do}(x), w) = P(y \mid \mathrm{do}(x))$ using Rule 1.

*Solution.* Rule 1 requires $Y \perp\!\!\!\perp_d W \mid X$ in $G_{\overline{X}}$ (delete edges into $X$; here $X$ has none, so $G_{\overline{X}} = G$). In $G$, the only path between $W$ and $Y$ is $W \leftarrow X \to Y$, a fork with fork-node $X$; since $X$ is in the conditioning set, this path is blocked, and there is no other path (by assumption $W, Y$ share no direct edge), so $Y \perp\!\!\!\perp_d W \mid X$ holds. Rule 1 therefore licenses dropping $w$ from the conditioning set: $P(y \mid \mathrm{do}(x), w) = P(y \mid \mathrm{do}(x))$. Practically, this says an irrelevant post-treatment side effect of $X$ — one that does not itself influence $Y$ — can always be safely ignored when computing $Y$'s response to the intervention, without needing to measure or adjust for it at all.

PART III · DO-CALCULUS

## 14.7 Instrumental Variables

Back-door needs the confounders, front-door needs a clean mediator. A third classical route uses a variable that moves $X$ *as if at random*.

**Definition 14.6** (Instrument)**.** $Z$ is an *instrumental variable* for the effect of $X$ on $Y$ if (a) *relevance*: $Z$ and $X$ are dependent; (b) *exclusion*: every directed path from $Z$ to $Y$ passes through $X$; (c) *exogeneity*: $Z$ shares no unblocked back-door path with $Y$ (e.g. $Z \perp\!\!\!\perp U$ for the hidden confounder $U$ of $X$ and $Y$).

Without further assumptions an instrument does *not* point-identify $P(y \mid \mathrm{do}(x))$ (the bow arc between $X$ and $Y$ remains; only bounds follow). Linearity (below) or monotonicity of $X$ in $Z$ restores identification; under monotonicity the identified quantity is the *local average treatment effect* (LATE) for the "compliers", the units whose treatment is moved by the instrument (Imbens & Angrist 1994).

**Proposition 14.3** (IV identification in the linear model)**.** *Let* $X = \alpha Z + \delta U + \varepsilon_X$ *and* $Y = \beta X + \gamma U + \varepsilon_Y$ *with* $Z, U, \varepsilon_X, \varepsilon_Y$ *mutually independent and first-stage coefficient* $\alpha \neq 0$. *Then*

$$\beta = \frac{\mathrm{Cov}(Z,Y)}{\mathrm{Cov}(Z,X)}, \qquad \text{and for binary } Z: \quad \beta = \frac{\mathbb{E}[Y \mid Z{=}1] - \mathbb{E}[Y \mid Z{=}0]}{\mathbb{E}[X \mid Z{=}1] - \mathbb{E}[X \mid Z{=}0]} \quad \textit{(Wald)}. \tag{14.6}$$

*The OLS slope instead tends to* $\beta + \gamma\delta\mathrm{Var}(U)/\mathrm{Var}(X)$.

*Proof.* By bilinearity and independence, $\mathrm{Cov}(Z,Y) = \beta\mathrm{Cov}(Z,X) + \gamma\mathrm{Cov}(Z,U) + \mathrm{Cov}(Z,\varepsilon_Y) = \beta\mathrm{Cov}(Z,X)$ and $\mathrm{Cov}(Z,X) = \alpha\mathrm{Var}(Z) \neq 0$. For binary $Z$ with $p = P(Z{=}1)$, $\mathrm{Cov}(Z,V) = p(1-p)\big(\mathbb{E}[V \mid Z{=}1] - \mathbb{E}[V \mid Z{=}0]\big)$ for any $V$, and the factor $p(1-p)$ cancels. For OLS, $\mathrm{Cov}(X,Y) = \beta\mathrm{Var}X + \gamma\mathrm{Cov}(X,U) = \beta\mathrm{Var}X + \gamma\delta\mathrm{Var}U$; divide by $\mathrm{Var}X$. □

PART III · DO-CALCULUS

## 14.8 c-Components and the ID Algorithm in Outline

When neither criterion applies, the complete answer is algorithmic. Represent each hidden common cause by a bidirected edge, giving Session 13's *acyclic directed mixed graph* (ADMG).

**Definition 14.7** (c-component)**.** Two observed variables are in the same *c-component* (confounded component) if they are connected by a path consisting only of bidirected edges. The c-components $S_1, \dots, S_k$ partition $\mathbf{V}$. For $S \subseteq \mathbf{V}$ define $Q[S](\mathbf{v}) = P(s \mid \mathrm{do}(\mathbf{v} \setminus s))$, the joint law of $S$ when everything else is set by intervention.

**Theorem 14.4** (c-component factorization; Tian & Pearl 2002)**.** $P(\mathbf{v}) = \prod_{j=1}^{k} Q[S_j]$, *and every factor is identified: with a topological order* $V_1 < \cdots < V_n$,

$$Q[S_j] = \prod_{i:\, V_i \in S_j} P(v_i \mid v_1, \dots, v_{i-1}).$$

*Moreover, if $A \subseteq C$ is ancestral in the subgraph induced by $C$, then $Q[A] = \sum_{c \setminus a} Q[C]$.*

**The ID algorithm (Shpitser & Pearl 2006), in outline.** To compute $P(y \mid \mathrm{do}(x))$: (1) let $D$ be the set of ancestors of $Y$ (including $Y$) in the graph $G[\mathbf{V} \setminus X]$ obtained by deleting $X$; then $P(y \mid \mathrm{do}(x)) = \sum_{d \setminus y} Q[D]$. (2) Split $D$ into its c-components $D_1, \ldots, D_m$ in $G[D]$, so $Q[D] = \prod_j Q[D_j]$. (3) Compute each $Q[D_j]$ from the identified $Q[S]$ of the c-component $S \supseteq D_j$ of $G$, by repeatedly marginalizing ancestral subsets and re-factorizing into c-components (Theorem 14.4). The algorithm fails exactly when some step meets a *hedge* — two nested c-forests, the generalisation of the bow arc — and then the effect is provably not identifiable.

**Worked Example 14.6** (The front-door formula from c-components)**.** *(Analytical.)* In the front-door graph the c-components are $\{X, Y\}$ (via $X \leftrightarrow Y$) and $\{M\}$. With order $X < M < Y$: $Q[\{M\}] = P(m \mid x)$ and $Q[\{X, Y\}] = P(x)P(y \mid x, m)$. Step 1: deleting $X$ leaves $M \to Y$, so the ancestors of $Y$ are $D = \{M, Y\}$ and $P(y \mid \mathrm{do}(x)) = \sum_m Q[\{M, Y\}]$. Step 2: in $G[D]$ there is no bidirected edge, so $Q[\{M, Y\}] = Q[\{M\}]\, Q[\{Y\}]$. Step 3: $\{Y\}$ is ancestral within $G[\{X, Y\}]$ (which has no directed edge), so $Q[\{Y\}] = \sum_{x'} Q[\{X, Y\}] = \sum_{x'} P(x')P(y \mid x', m)$ (here $m$ is a fixed argument, since $M$ is intervened on in $Q[\{X, Y\}]$). Together: Eq. (14.5). In the bow-arc graph, by contrast, $D = \{Y\}$ lies in the c-component $\{X, Y\}$ but $\{Y\}$ is not ancestral there (the arrow $X \to Y$ is inside it): the hedge.

PART III · DO-CALCULUS

## 14.9 Non-Identifiability: When No Formula Exists

**Worked Example 14.7** (The bow-arc graph)**.** Let $U$ be an unobserved common cause of both $X$ and $Y$, with also a direct edge $X \to Y$, and no measured mediator at all. No set of *observed* variables can satisfy the back-door criterion (the only back-door path $X \leftarrow U \to Y$ passes through the unmeasured $U$, and there are no other measured variables to substitute), and the front-door criterion requires a mediator $M$ intercepting the $X \to Y$ edge, which does not exist here either.

**Theorem 14.5** (Non-identifiability of the bow-arc graph)**.** *For this bow-arc graph, no application of do-calculus (Theorem 14.3) can reduce $P(y \mid \mathrm{do}(x))$ to an expression in $P(x, y)$ alone; there exist two distinct SCMs sharing this graph and inducing the identical observational distribution $P(x, y)$ but different interventional distributions $P(y \mid \mathrm{do}(x))$.*

*Proof.* It suffices to construct the two models (then, by soundness of do-calculus, no derivation exists). Fix any strictly positive $P(x, y)$ on $\{0, 1\}^2$. *Model A:* $U = (U_X, U_Y)$ with independent components, $X := U_X$ with $P(U_X{=}1) = P(X{=}1)$, and $Y := \mathbb{1}\{U_Y < P(Y{=}1 \mid X)\}$ with $U_Y \sim \mathrm{Unif}(0, 1)$. It reproduces $P(x, y)$, and since $U_Y \perp\!\!\!\perp X$, $P_A(Y{=}1 \mid \mathrm{do}(x)) = P(Y{=}1 \mid x)$. *Model B:* $U = (U_1, U_2)$ drawn jointly from $P(x, y)$, $X := U_1$, $Y := U_2$. It also reproduces $P(x, y)$, but $Y$ does not listen to $X$, so $P_B(Y{=}1 \mid \mathrm{do}(x)) = P(Y{=}1)$ for both $x$. Both models are compatible with the bow arc (an SCM may leave an arrow of the graph unused). The two interventional distributions coincide only if $P(Y{=}1 \mid x) = P(Y{=}1)$, i.e. if $X \perp\!\!\!\perp Y$; whenever the data show any dependence, they differ. □

**Worked Example 14.8** (Two SCMs, one dataset). *(Numerical.)* Take $P(X{=}1) = 0.5$, $P(Y{=}1 \mid X{=}1) = 0.8$, $P(Y{=}1 \mid X{=}0) = 0.3$, i.e. $P(x,y) = P(x)P(y \mid x) = (0.5 \cdot 0.7, 0.5 \cdot 0.3, 0.5 \cdot 0.2, 0.5 \cdot 0.8) = (0.35, 0.15, 0.10, 0.40)$ for $(x,y) = (0,0), (0,1), (1,0), (1,1)$. Model A gives $P(Y{=}1 \mid \mathrm{do}(x)) = P(Y{=}1 \mid x)$: 0.80 and 0.30, an ATE of $0.80 - 0.30 = 0.50$. Model B gives $P(Y{=}1 \mid \mathrm{do}(x)) = P(Y{=}1) = 0.15 + 0.40 = 0.55$ for both $x$, an ATE of 0. The data cannot tell a strong effect from no effect at all.

**PYTHON LAB** · s14_examples.py · ex_nonident()

```
Model A: P(Y=1|do(1)) = 0.80, P(Y=1|do(0)) = 0.30, ATE = 0.50
Model B: P(Y=1|do(1)) = 0.55, P(Y=1|do(0)) = 0.55, ATE = 0.00
bounds:  P(Y=1|do(1)) in [0.40, 0.90], P(Y=1|do(0)) in [0.15, 0.65]; ATE in
[-0.25, 0.75]
```

This is not a failure of cleverness — it is a mathematical impossibility, and recognizing it (rather than reporting a confidently wrong number) is itself the valuable skill this session teaches. When a causal graph implies non-identifiability, the only paths forward are: collect more data (measure a proxy for $U$, breaking the bow), run a randomized experiment (Session 15's benchmark case, which sidesteps identification entirely by design), or bound the effect rather than pin it down exactly (partial identification: for a binary outcome, $P(Y{=}1, X{=}x) \le P(Y{=}1 \mid \mathrm{do}(x)) \le P(Y{=}1, X{=}x) + 1 - P(X{=}x)$ without any assumption; Manski 1990).

PART III · DO-CALCULUS

## 14.10 From Question to Estimand: A Practical Recipe

1 *Question:* write the target as $P(y \mid \mathrm{do}(x))$, an ATE or a policy value $P(y \mid \mathrm{do}(\pi))$. 2 *Graph:* draw the causal DAG with domain experts, unmeasured variables included; every missing arrow is an assumption. 3 *Identify*: back-door, front-door, an instrument, then the ID algorithm; if all fail, randomize, measure a proxy, bound, or change the question. 4 *Estimate* the functional (Part IV, Session 15). 5 *Stress-test* testable implications (Session 13), positivity and hidden confounding.

*Remark* 14.1 (Sensitivity analysis and the E-value). How strong would an unmeasured confounder have to be to explain an observed risk ratio $RR > 1$ away? VanderWeele & Ding (2017): its risk-ratio associations with both treatment and outcome must be at least

$$E = RR + \sqrt{RR\,(RR-1)} \qquad (\text{use } 1/RR \text{ if } RR < 1). \tag{14.7}$$

For the kidney data the adjusted risk ratio is $RR = 0.83255/0.77887 = 1.0689$ (unrounded values of Worked Example 14.1), so $E = 1.0689 + \sqrt{1.0689 \times 0.0689} = 1.0689 + \sqrt{0.0737} = 1.0689 + 0.2714 = 1.340$: a modest confounder (for instance a second severity dimension not captured by stone size) could erase the advantage of A. For comparison, $RR = 1.5$ gives $E = 1.5 + \sqrt{0.75} = 2.366$ and $RR = 1.8$ gives $E = 1.8 + \sqrt{1.8 \times 0.8} = 1.8 + 1.2 = 3.000$.

**PYTHON LAB** · s14_examples.py · ex_evalue()

```
kidney adjusted risk ratio RR = 0.83255/0.77887 = 1.0689; E-value = 1.0689 +
0.2714 = 1.340
```

```
RR = 1.5 -> E-value = 2.366; RR = 1.8 -> E-value = 3.000
```

PART IV

# Identification in Agents, RL and Multi-Agent Systems

Every action is an intervention; every log is observational data

PART IV · AI CONNECTIONS

## 14.11 Reinforcement Learning: Actions Are Interventions

An RL agent does not observe its actions — it *sets* them. With $G_t = \sum_{k\geq 0} \gamma^k R_{t+k+1}$ the discounted return ($0 \leq \gamma < 1$), the action value is an interventional quantity,

$$Q^\pi(s,a) = \mathbb{E}\big[G_t \mid S_t = s,\ \mathrm{do}(A_t = a),\ \text{then follow } \pi\big], \qquad V^\pi = \mathbb{E}\big[G_t \mid \mathrm{do}(\pi)\big],$$

and the value of a policy is exactly the policy intervention of Proposition 14.1. Online, the agent's own randomization makes conditioning and doing coincide. Offline, the data come from a behaviour (logging) policy $\mu(a \mid s)$, and identification is required; estimating a target policy's value from such logs is *off-policy evaluation* (OPE): if $\mu$ chose actions using only the recorded state, $S$ is a back-door set for $(A, R)$ and both the adjustment ("direct method") and the IPW ("importance sampling") forms of Proposition 14.2 apply.

**Worked Example 14.9** (Off-policy evaluation as identification)**.** *(Numerical, hypothetical bandit.)* Context $S \in \{0,1\}$ with $P(S{=}1) = 0.3$; the logging policy plays $A = 1$ with probability $\mu(1 \mid 0) = 0.2$ in $s = 0$ and $\mu(1 \mid 1) = 0.9$ in $s = 1$. Mean rewards $r(s,a) = \mathbb{E}[R \mid s, a]$: $r(0,0) = 1$, $r(0,1) = 2$, $r(1,0) = 0$, $r(1,1) = 0.5$. The target policy "always $a = 1$" has value $\sum_s P(s) r(s,1) = 0.7(2) + 0.3(0.5) = 1.4 + 0.15 = 1.55$. The naive logged average weights the contexts by $P(s \mid A{=}1)$: $P(S{=}0, A{=}1) = 0.7(0.2) = 0.14$ and $P(S{=}1, A{=}1) = 0.3(0.9) = 0.27$, so $\mathbb{E}[R \mid A{=}1] = (0.14(2) + 0.27(0.5))/0.41 = 0.415/0.41 = 1.012$, dominated by the poor context $s = 1$, where $a = 1$ was logged most. *Simulation* (`ex_rl_ope()`): $n = 20\,000$ logged rounds, rewards $r(s,a)$ plus Gaussian noise with sd 0.5, seed 17. Importance sampling $\frac{1}{n}\sum_i \mathbb{1}(A_i{=}1) R_i / \mu(1 \mid S_i) = 1.568$; adjustment ("direct method") $\sum_s \hat{P}(s)\hat{Q}(s,1) = 1.549$, where $\hat{P}(s)$ is the logged frequency of context $s$ and $\hat{Q}(s,1)$ the average logged reward over rounds with $S = s$, $A = 1$.

**PYTHON LAB** · `s14_examples.py` · `ex_rl_ope()`

```
P(S=0,A=1) = 0.14, P(S=1,A=1) = 0.27; naive = 0.415/0.41
true V(always a=1) = 1.550; naive E[R|A=1] = 1.012
logged n=20000, seed=17:  importance sampling = 1.568; direct (adjustment) =
1.549
```

PART IV · AI CONNECTIONS

## 14.12 Agentic AI: What If the Agent Always Called the Tool?

**Worked Example 14.10** (Can the logs answer the product question?). *(Numerical, hypothetical.)* An agent's past policy called a retrieval tool $T$ on 80% of hard queries and 30% of easy ones; 40% of queries are hard (query difficulty $Z$). Success rates: hard 0.6 with $T$, 0.3 without; easy 0.9 with $T$, 0.85 without. The joint frequencies are $P(T, \text{hard}) = 0.4(0.8) = 0.32$, $P(T, \text{easy}) = 0.6(0.3) = 0.18$, $P(\text{no } T, \text{hard}) = 0.08$, $P(\text{no } T, \text{easy}) = 0.42$, so half of all logged queries used $T$. The naive success rates are $P(\text{succ} \mid T) = (0.32(0.6) + 0.18(0.9))/0.5 = 0.354/0.5 = 0.708$ and $P(\text{succ} \mid \text{no } T) = (0.08(0.3) + 0.42(0.85))/0.5 = 0.381/0.5 = 0.762$: "the tool hurts" $(-0.054)$. Difficulty is recorded and is the only common cause, so it is a back-door set: $P(\text{succ} \mid \text{do}(T)) = 0.4(0.6) + 0.6(0.9) = 0.24 + 0.54 = 0.78$ and $P(\text{succ} \mid \text{do}(\text{no } T)) = 0.4(0.3) + 0.6(0.85) = 0.12 + 0.51 = 0.63$, an effect of $+0.15$. The conditional policy "call $T$ only on hard queries" (Eq. (14.2)) reaches $0.4(0.6) + 0.6(0.85) = 0.75$ with only 0.40 tool calls per query — a cost–benefit trade-off computed entirely from logs.

**PYTHON LAB** · s14_examples.py · ex_agent_tool()

```
P(T) = 0.50; numerators:  T 0.192 + 0.162 = 0.354, noT 0.024 + 0.357 = 0.381
naive:  P(succ|T) = 0.708, P(succ|noT) = 0.762, diff = -0.054
adjusted:  P(succ|do(T)) = 0.78, P(succ|do(noT)) = 0.63, effect = +0.15;
policy 'T only on hard' 0.75
```

PART IV · AI CONNECTIONS

## 14.13 Multi-Agent AI: Interventions Ripple Through Other Agents

Intervening on one agent changes the inputs of the others, who may adapt; "the effect of agent 1" is ambiguous until we say what happens to everyone else.

**Worked Example 14.11** (Total versus controlled direct effect of an agent). *(Numerical, hypothetical.)* Agent 1 chooses $X \in \{0, 1\}$; agent 2 observes it and plays $A_2 = 1$ with probability 0.7 if $X = 1$ and 0.2 if $X = 0$; the team reward has $\mathbb{E}[Y \mid x, a_2] = 0.2, 0.5, 0.4, 0.9$ for $(x, a_2) = (0,0), (0,1), (1,0), (1,1)$. Nothing else influences $X$, so $\mathbb{E}[Y \mid \text{do}(x)] = \sum_{a_2} P(a_2 \mid x)\mathbb{E}[Y \mid x, a_2]$. The *total* effect with agent 2 adapting is therefore $\mathbb{E}[Y \mid \text{do}(X{=}1)] - \mathbb{E}[Y \mid \text{do}(X{=}0)] = \big(0.3(0.4) + 0.7(0.9)\big) - \big(0.8(0.2) + 0.2(0.5)\big) = (0.12 + 0.63) - (0.16 + 0.10) = 0.75 - 0.26 = 0.49$. Freezing agent 2 by a second intervention gives *controlled direct* effects $\mathbb{E}[Y \mid x{=}1, a_2] - \mathbb{E}[Y \mid x{=}0, a_2]$: $0.4 - 0.2 = 0.20$ under $\text{do}(A_2{=}0)$ and $0.9 - 0.5 = 0.40$ under $\text{do}(A_2{=}1)$. Three legitimate numbers answer three different questions; identifying the direct effects from logs requires that nothing unobserved confounds $A_2$ and $Y$ given $X$ — in a game, shared private signals often do.

**PYTHON LAB** · s14_examples.py · ex_multiagent()

```
E[Y|do(X=1)] = 0.75, E[Y|do(X=0)] = 0.26, total effect = 0.49
controlled direct effect with do(A2=0):  0.20; with do(A2=1):  0.40
```

**AI CONNECTION · Generative Models and AI Paradigms**

A deep generative model fitted to observational data (course session 4) learns $P(\mathbf{v})$ — rung 1 of the ladder of causation (Session 12); even a perfect VAE or diffusion model cannot answer $P(y \mid \mathrm{do}(x))$ in a non-identified graph (Theorem 14.5 holds for any model class). Do-calculus is symbolic inference over graphs (course session 2); pairing it with neural estimators is neuro-symbolic AI in miniature: symbols decide *what* to estimate, networks estimate it.

PART IV · AI CONNECTIONS

## 14.14 Summary

**THINK ABOUT IT** **A hospital argues that because its database has two million patients, the effect of a drug is "certainly identified". What is wrong with this argument, and what would you ask for instead?**

**Table 14.3.** Cheat sheet of key formulas.

| Object | Formula / condition |
|---|---|
| **Truncated factorization** | $P(\mathbf{v} \mid \mathrm{do}(x)) = \prod_{i \notin X} P(v_i \mid \mathrm{pa}_i)\big\vert_{X=x}$ |
| **Back-door criterion** | $Z \cap \mathrm{de}(X) = \emptyset$ and $X \perp\!\!\!\perp_d Y \mid Z$ in $G_{\underline{X}}$ |
| **Back-door adjustment** | $P(y \mid \mathrm{do}(x)) = \sum_z P(y \mid x, z)P(z) = \mathbb{E}\big[\mathbb{1}(X{=}x)\mathbb{1}(Y{=}y)/P(x \mid Z)\big]$ |
| **Policy value** | $P(y \mid \mathrm{do}(\pi)) = \sum_{z,x} P(y \mid x, z)\pi(x \mid z)P(z)$ |
| **Front-door** | $P(y \mid \mathrm{do}(x)) = \sum_m P(m \mid x) \sum_{x'} P(y \mid x', m)P(x')$ |
| **Do-calculus** | Rule 1: $G_{\overline{X}}$; Rule 2: $G_{\overline{X}\underline{Z}}$; Rule 3: $G_{\overline{X}\,\overline{Z(W)}}$ |
| **IV (linear)** | $\beta = \mathrm{Cov}(Z, Y)/\mathrm{Cov}(Z, X)$; exclusion failure adds $\theta/\alpha$ |
| **Plug-in variance** | $V = \sum_z p_z \big[\frac{\mu_{1z}(1-\mu_{1z})}{e_z} + \frac{\mu_{0z}(1-\mu_{0z})}{1-e_z} + (\tau_z - \tau)^2\big]$ |
| **Bounds; E-value** | $P(X{=}x, Y{=}1) \le P(Y{=}1 \mid \mathrm{do}(x)) \le P(X{=}x, Y{=}1) + P(X{\neq}x)$; $E = RR + \sqrt{RR(RR-1)}$ |

**Further reading.** Pearl (2009), *Causality*, Ch. 3–4; Pearl, Glymour & Jewell (2016), *Causal Inference in Statistics: A Primer*, Ch. 3; Hernán & Robins (2020), *Causal Inference: What If*, Ch. 7, 12, 16; Peters, Janzing & Schölkopf (2017), Ch. 6; Shpitser & Pearl (2006) on the ID algorithm; Wasserman (2004), *All of Statistics*, Ch. 8.

**CONNECTIONS AND TAKEAWAYS**

We can now determine, directly from a graph, *whether* $P(y \mid \mathrm{do}(x))$ is a well-defined function of observational data (back-door, front-door, instruments with extra assumptions, or in general via do-calculus and the ID algorithm) and, when it is, write that function down explicitly; when it is not, we can bound it and measure how sensitive a conclusion is to hidden confounding. This is a complete answer to the identification question posed at the top of this session.

What we have *not* done systematically is estimate these quantities from finite, noisy, high-dimensional data — apart from the plug-in, bootstrap, IV and importance-sampling calculations of Parts IV and V, every formula above was applied either to population-level quantities or to a dataset treated as if it perfectly revealed $P(y \mid x, z)$ and $P(z)$. **Session 15**, the final session of these notes, closes this last gap: given real observational data, how do we actually build good estimators of the identified causal effect — propensity scores, inverse propensity weighting, doubly robust estimation, and machine-learning-based treatment-effect estimators — and how does the special case of a randomized experiment (where identification is guaranteed by design) connect back to everything built since Session 12?